



05

Google Kubernetes Engine Logging and Monitoring

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- 01 Google Cloud Observability
- 02 Cloud Logging
- 03 Inspecting logs with the kubectl command
- 04 Inspecting logs with Cloud Logging and Logging Agents
- 05 Cloud Monitoring

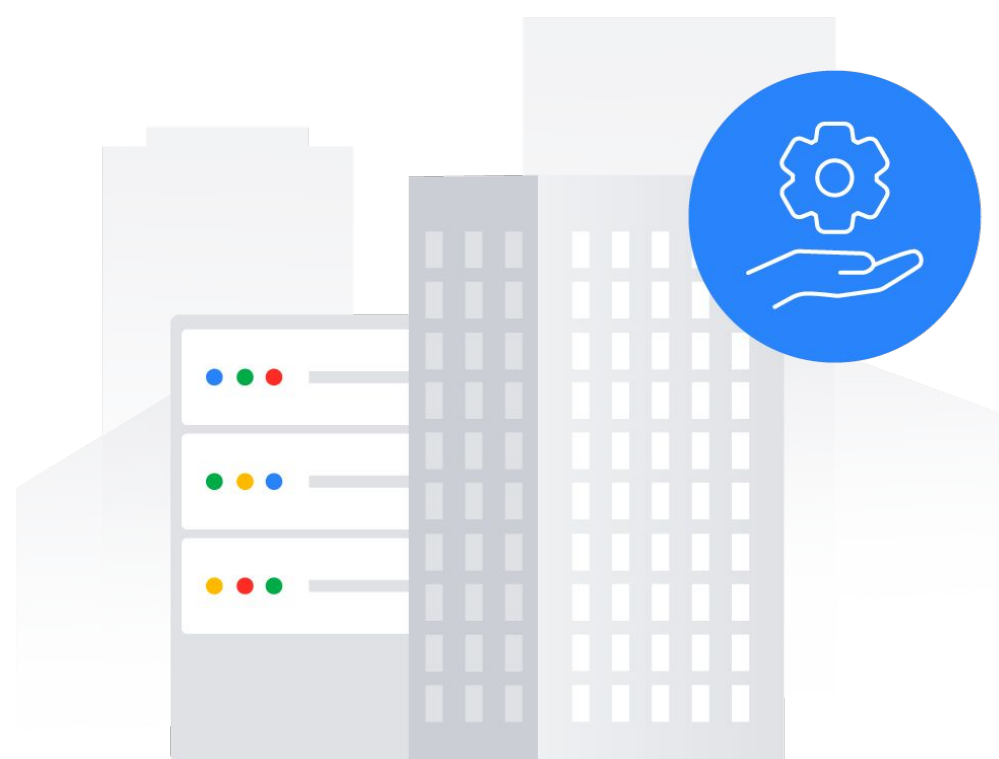


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Observability on-premises versus in the cloud



On-premises

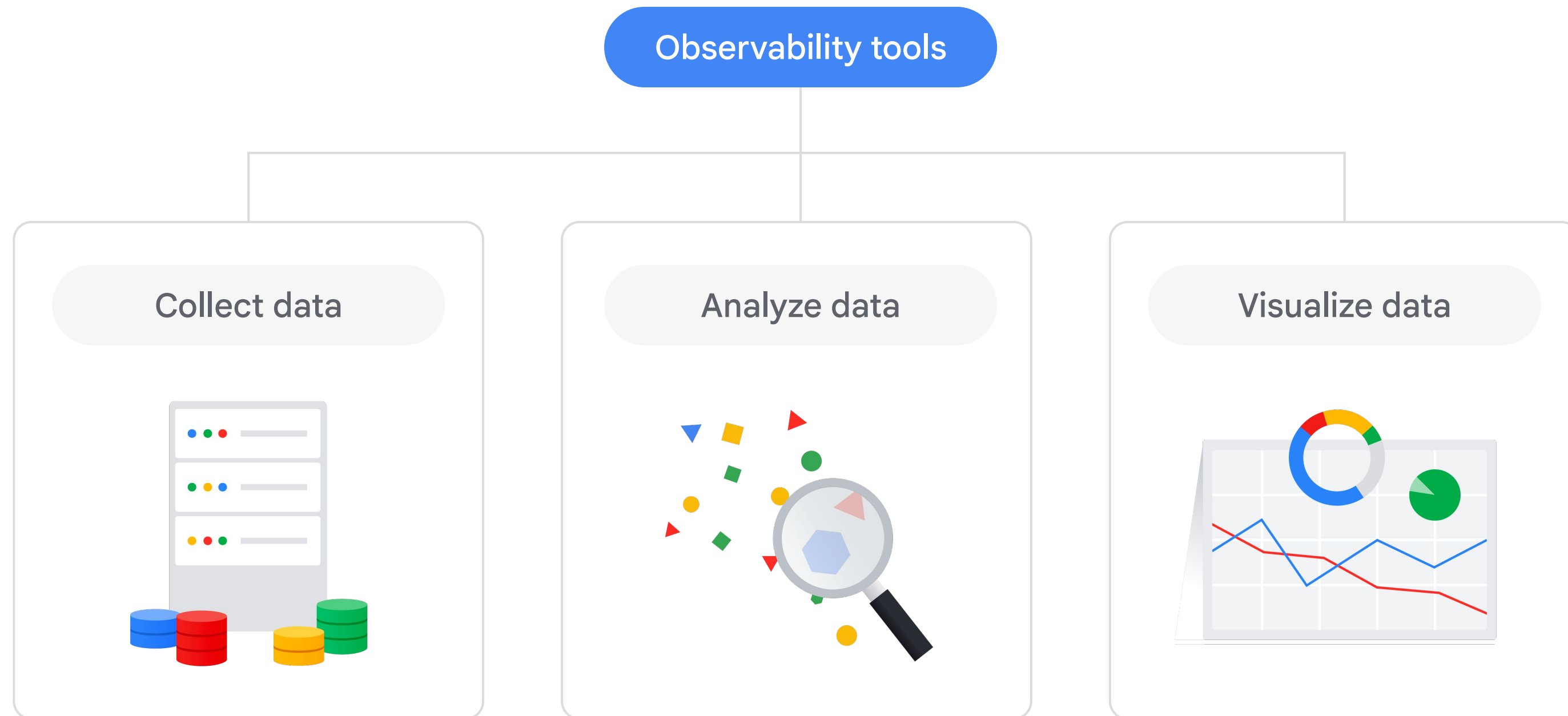
Physical check



Cloud

Observability tools

How can you know what's happening with your apps?



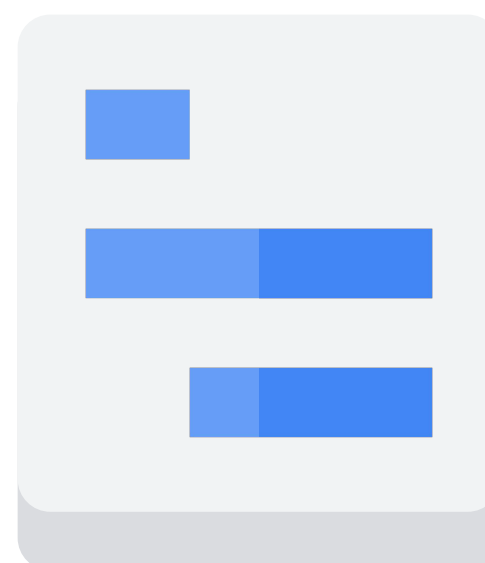
Google Cloud Observability



Cloud
Monitoring



Cloud
Logging



Cloud
Trace

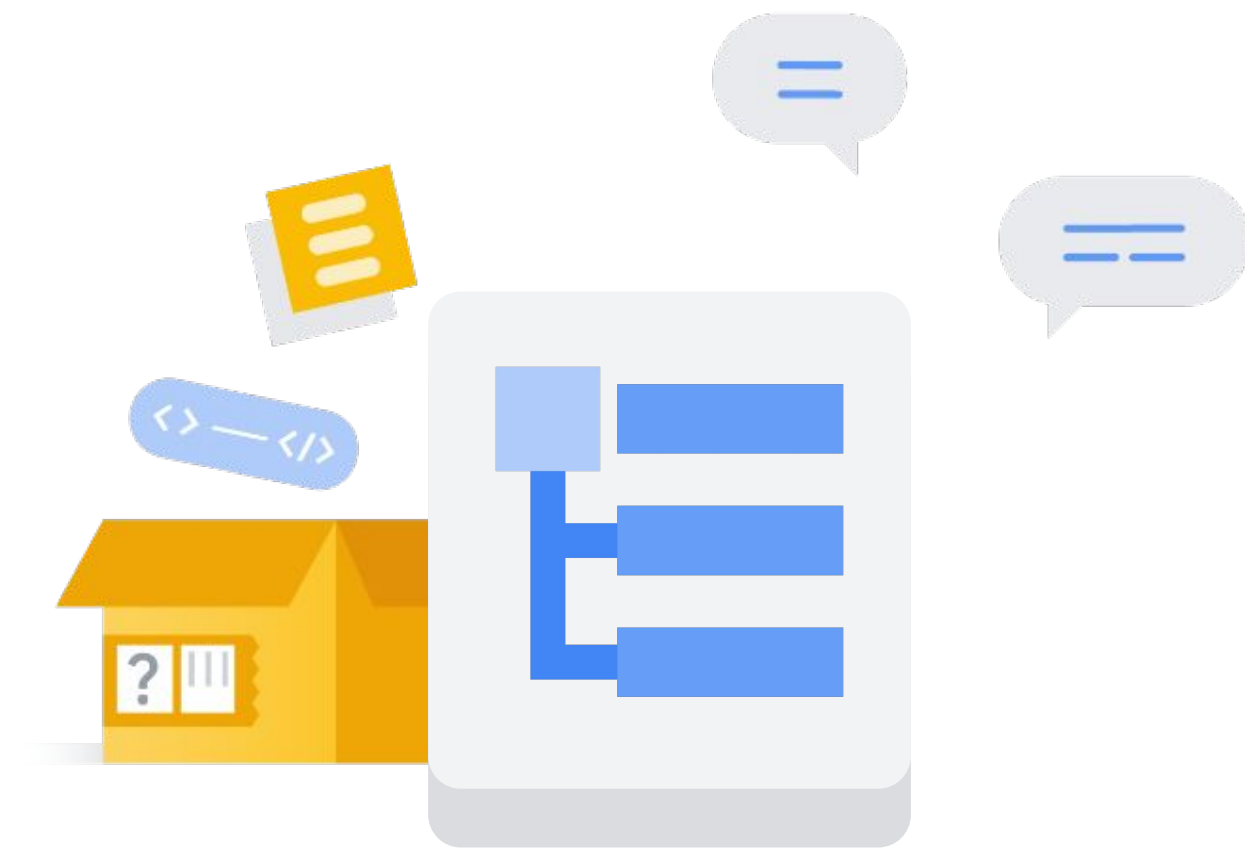


Cloud
Profiler

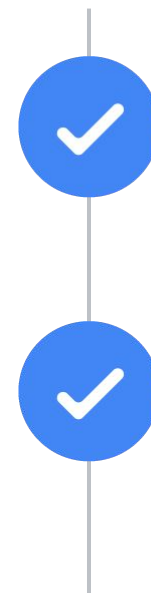


Error
Reporting

Cloud Logging



Cloud
Logging



Collects and stores all application and infrastructure logs.

Real-time insights help troubleshoot issues, identify trends, and comply with regulations.

Cloud Monitoring



Cloud
Monitoring



Provides a comprehensive view of cloud infrastructure and applications.

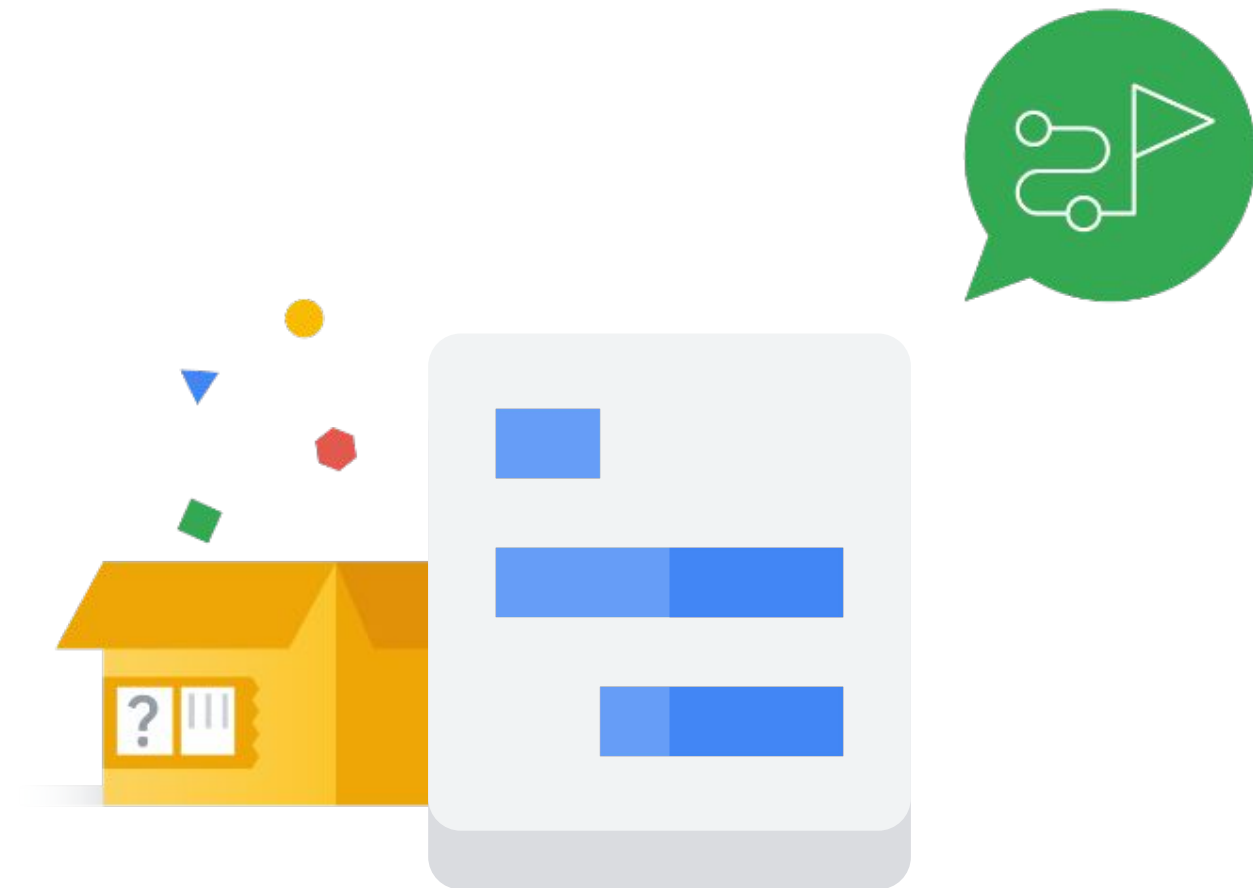


Collects metrics from applications and infrastructure, and provides insights into their performance, health, and availability.

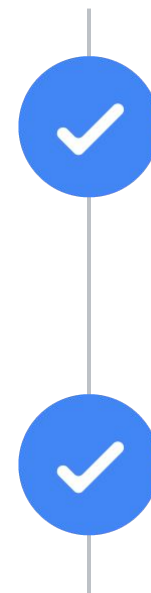


Lets you create alerting policies to notify when metrics, health check results, and uptime check results meet specified criteria.

Cloud Trace



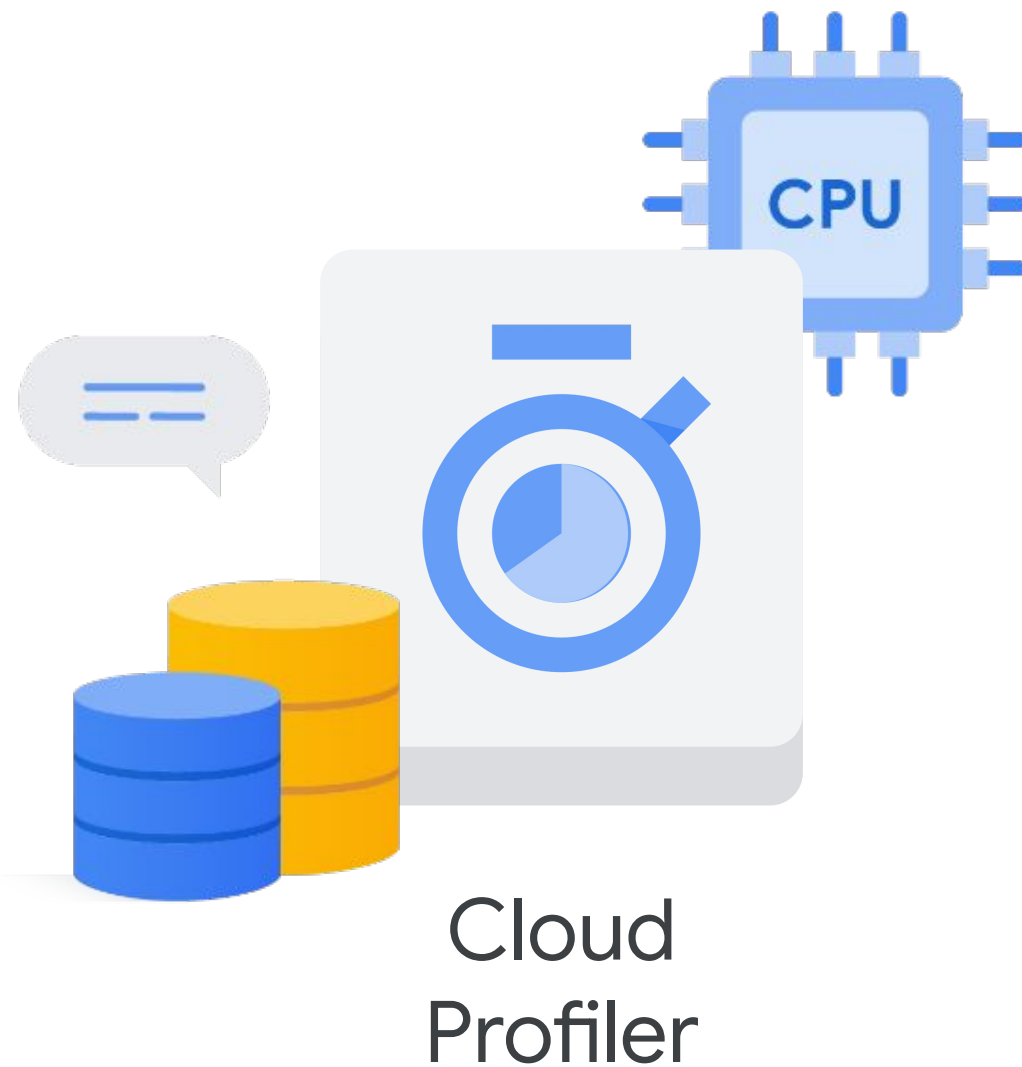
Cloud
Trace



Helps identify performance bottlenecks in applications.

Collects latency data from applications and provides insights into how they're performing.

Cloud Profiler



Identifies how much CPU power, memory, and other resources an application uses.



Continuously gathers CPU usage and memory-allocation information from production applications.



Provides insights into how applications are using resources.

Error Reporting



Error
Reporting



Error Reporting counts, analyzes, and aggregates the crashes in running cloud services in real time.



A centralized error management interface displays the results with sorting and filtering capabilities.



A dedicated view shows the error details: time chart, occurrences, affected user count, first- and last-seen dates.



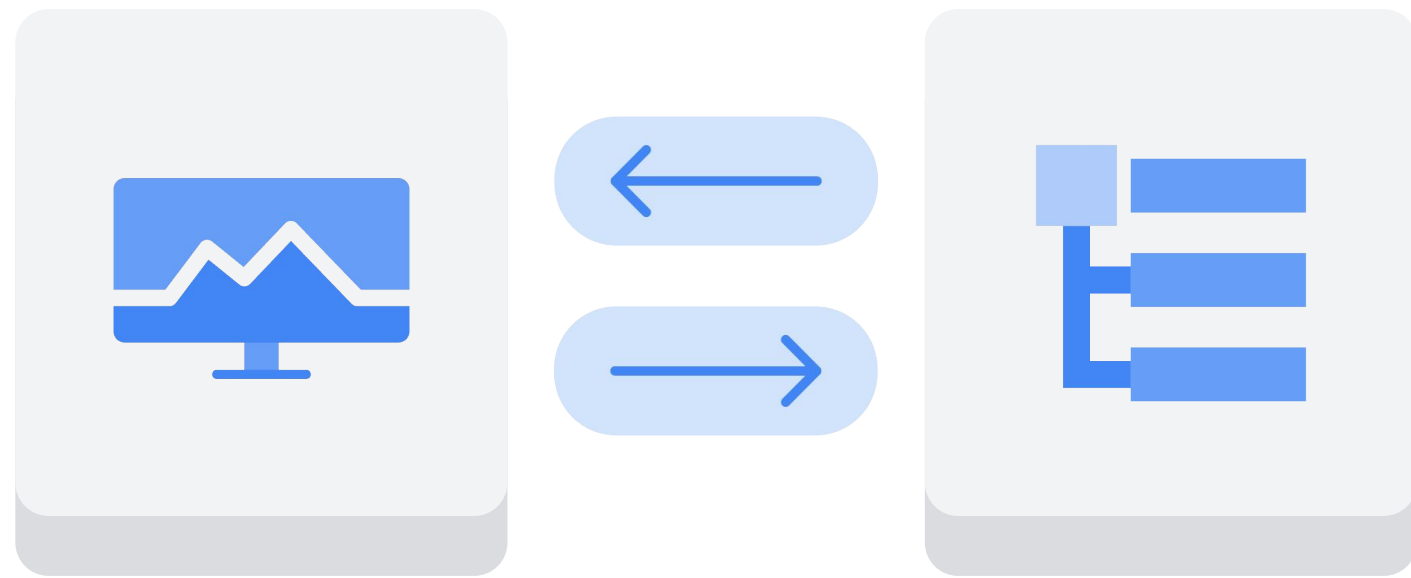
Error Reporting supports email and mobile notifications through its API.

Google Kubernetes Engine Logging and Monitoring

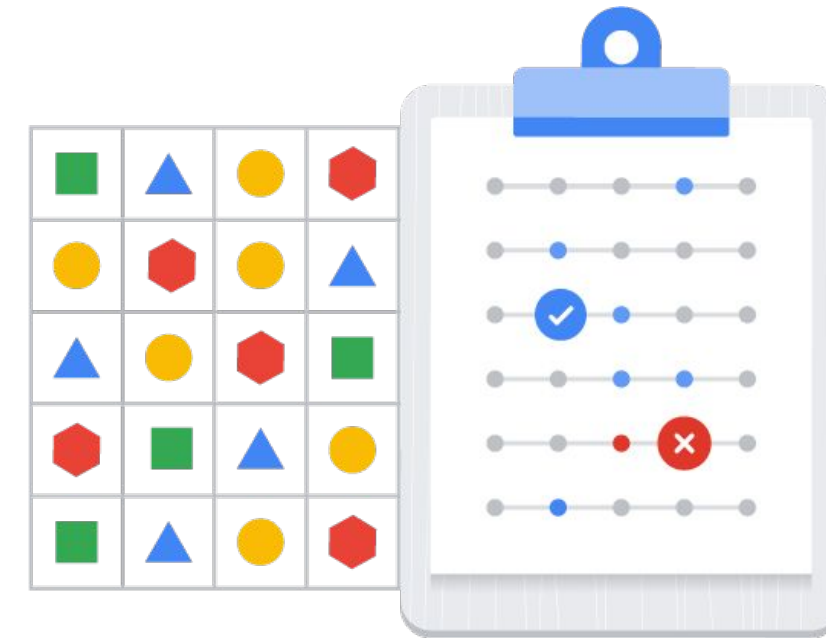
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Logging is a passive form of systems monitoring



The process of collecting and storing a record of events that occur in a GKE cluster.



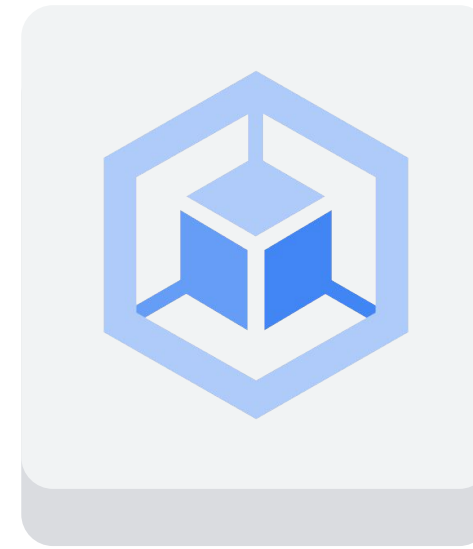
Example: collect events logs to identify patterns or the root cause of a system failure.

GKE Logs and Cloud Logging are different



Cloud Logging

Used to collect and analyze logs from a variety of sources.

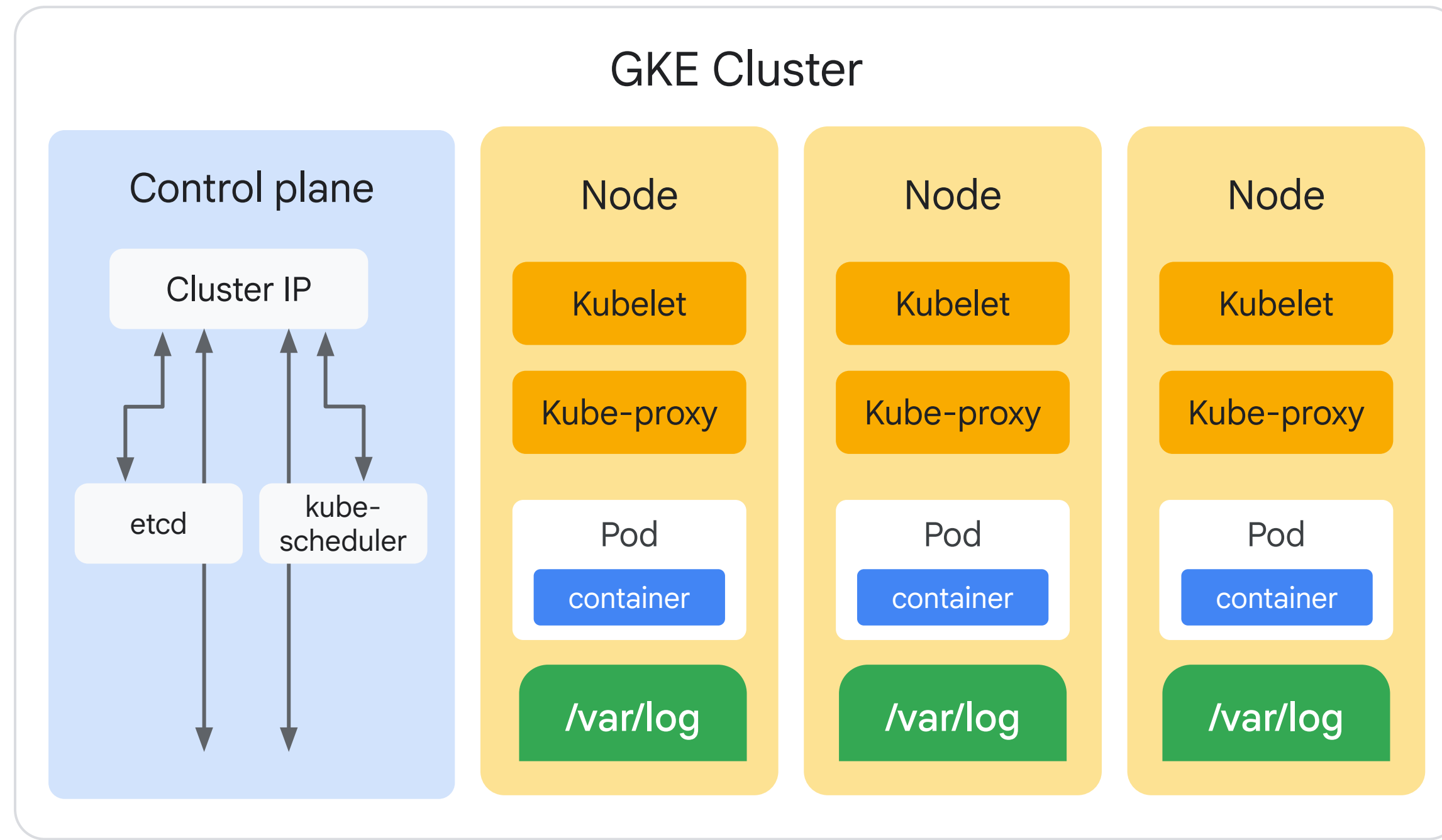


GKE Logs

A record of information about events and general systems details.

- Systems
- Workloads
- API server
- Scheduler
- Controller manager

Kubernetes logs are stored on node file systems



/var/log directory =

The default location for storing logs generated by applications and system processes that run on GKE nodes.

Not meant for long-term log storage.

For long-term storage, use Cloud Logging.

Options to view logs

1

kubectl logs command

```
$ kubectl logs [POD_Name]
```

Display logs directly from a Pod.



Quick, but logs aren't saved to any database.

2

Cloud Logging

Provides a single interface to review logs from containers, nodes, and other system services.



More options to broaden visibility of GKE events or correlate issues.

Cloud Logging storage and retention periods



Default log storage



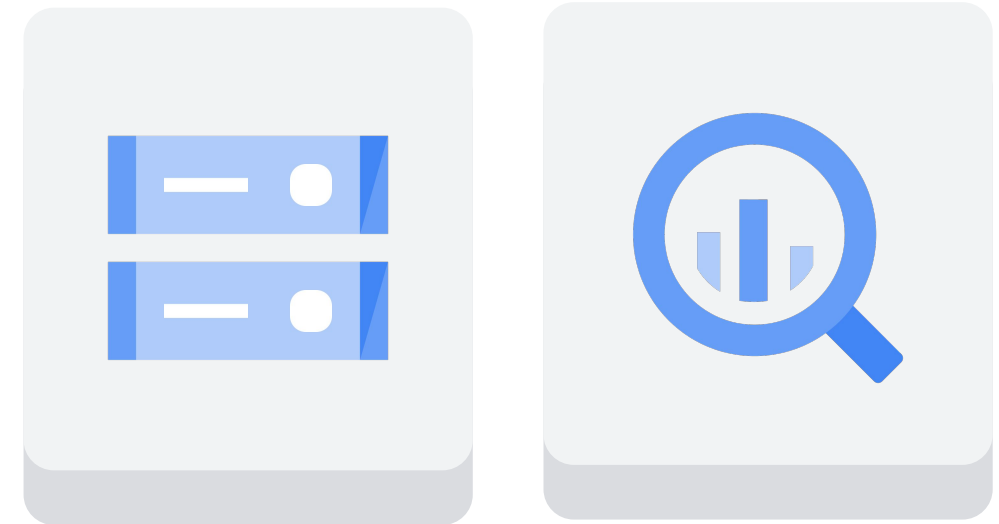
50GiB of logs free
for **30 days**



Required log storage



Up to **400 days**



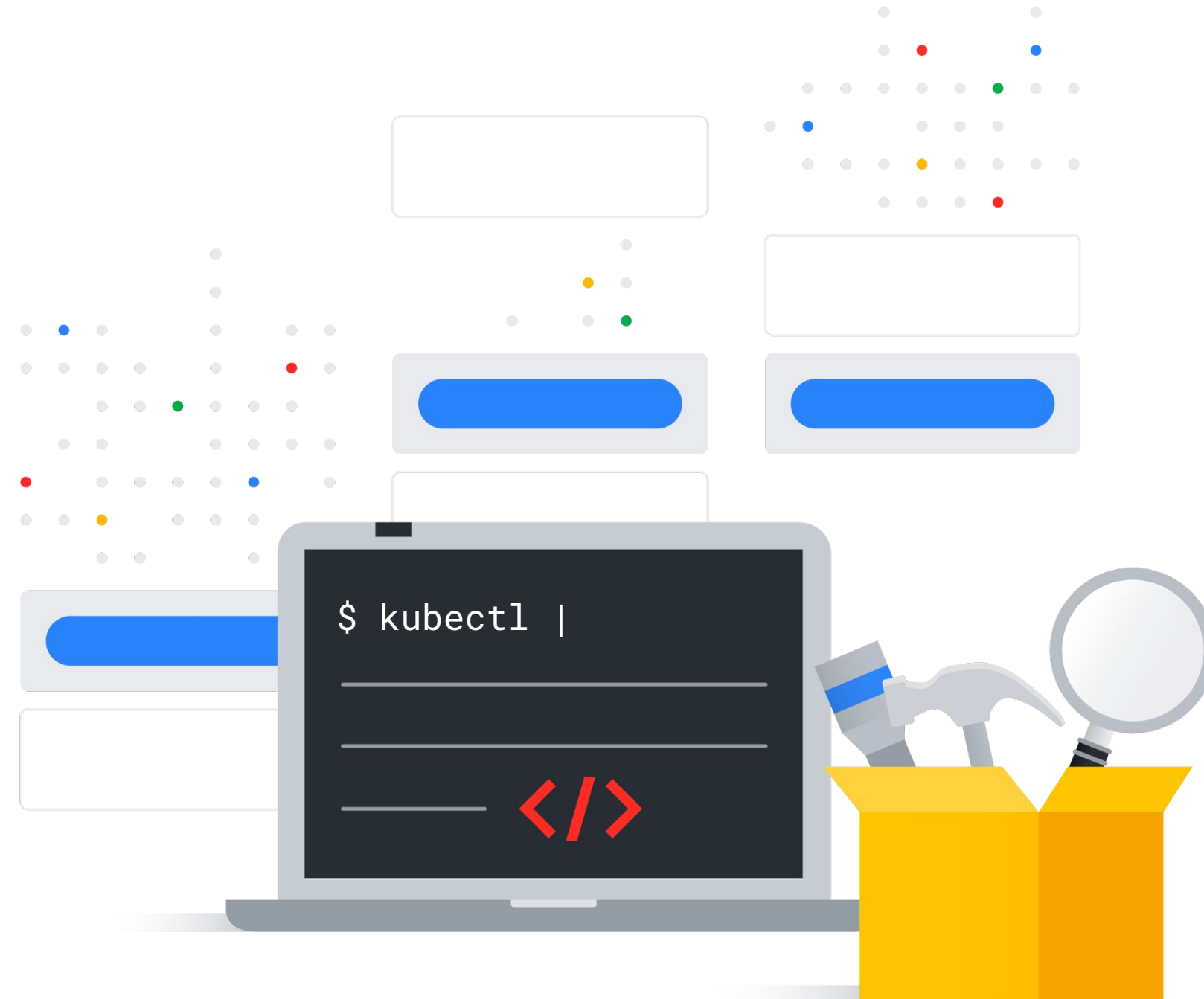
For longer retention
options, logs can be
exported to Cloud
Storage and BigQuery.

Google Kubernetes Engine Logging and Monitoring

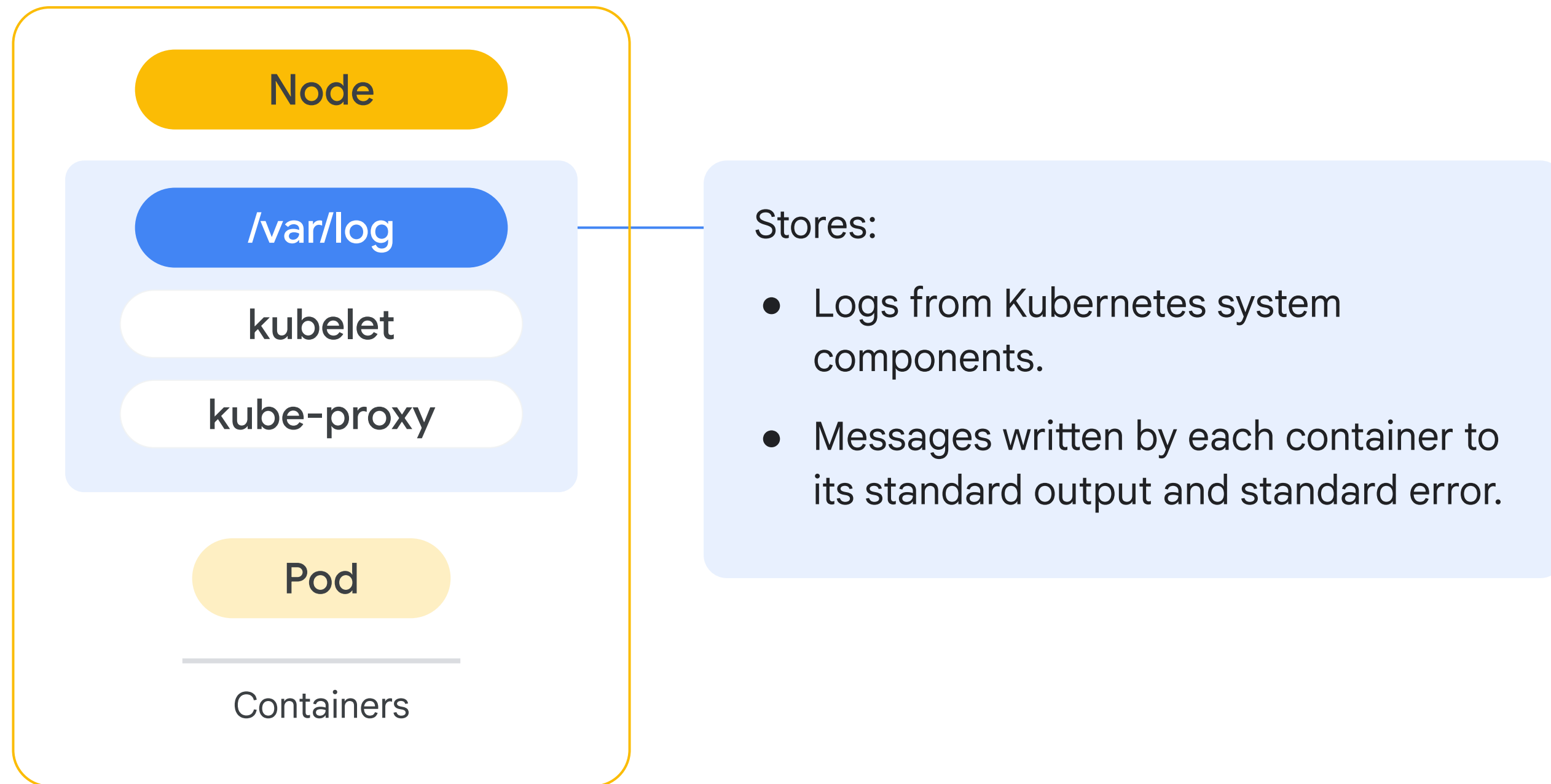
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Inspecting Logs with kubectl



Kubernetes system logs in /var/log



Producing logs with kubectl

```
$ kubectl logs [POD_NAME]
```

- Used to view the logs of a specific Pod.
- Run the `kubectl get pods` command to get a list of all the available Pods.

```
$ kubectl --tail=20 [POD_NAME]
```

- Used to limit the number of logs returned.
- By adding `--tail=20`, the output would be limited to the 20 most recent lines.

Producing logs with kubectl

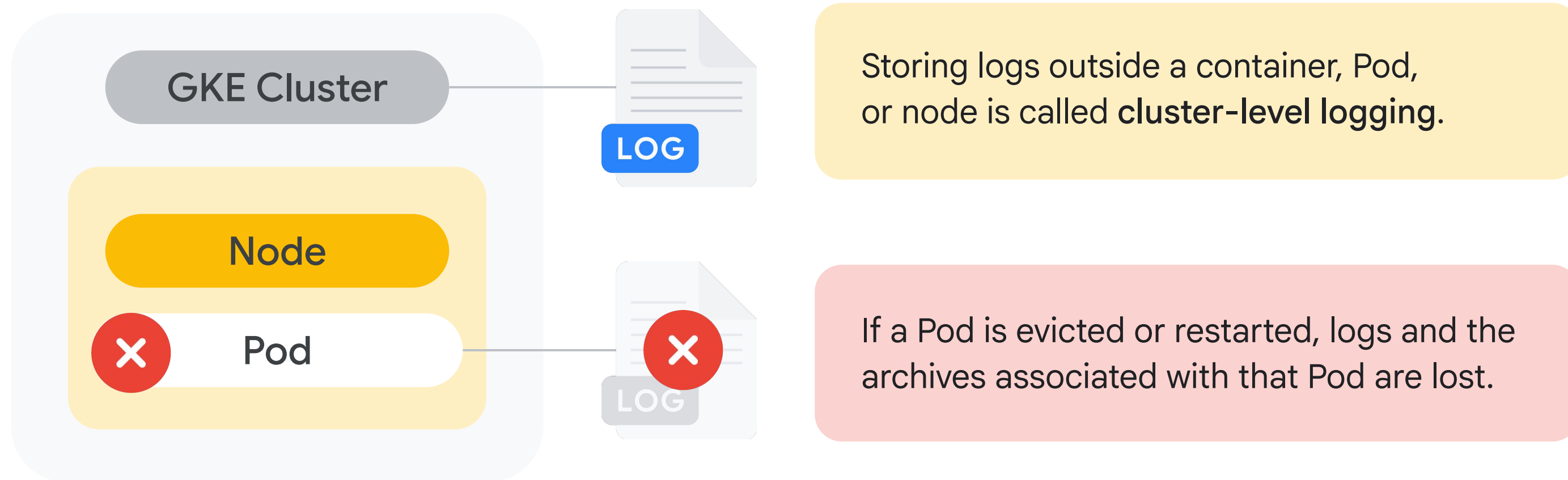
```
$ kubectl logs --since=3h [POD_NAME]
```

- Used to restrict the output based on time.
- `--since=3h` would return all logs for specific Pod or container from the last 3 hours.

```
$ kubectl logs --previous [POD_NAME]
```

- Used to view a previous instantiation of a container.
- If your Pod has multiple containers, specify which container's log the `kubectl` command should return.

Native Kubernetes Logs are not persistent



In GKE, log storage is handled by the integration with Cloud Logging.

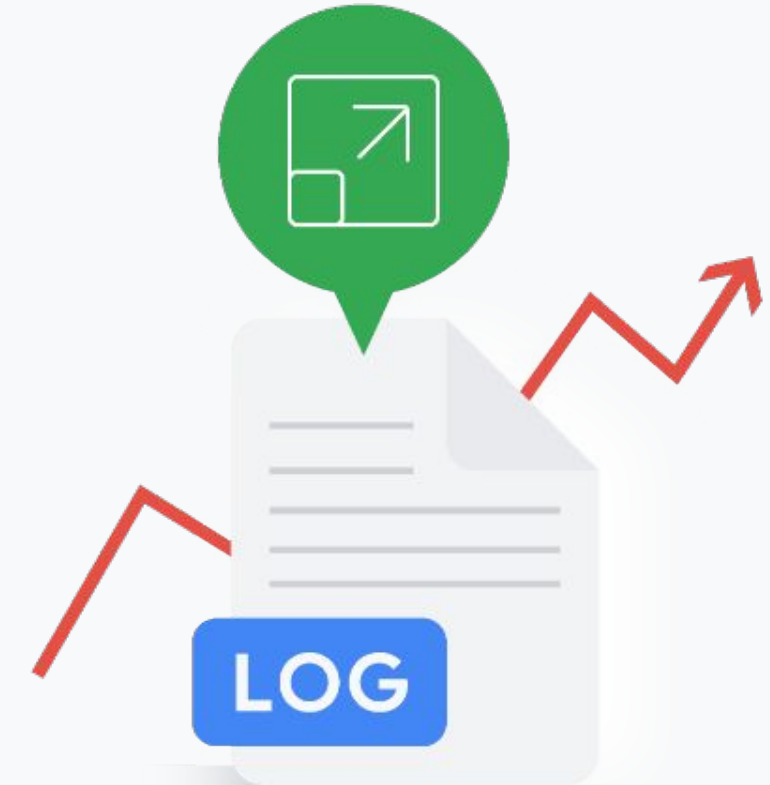
Creating a log file



A log file is created when a container starts on a node.



Node-level logging implements log archiving using a rotation mechanism.

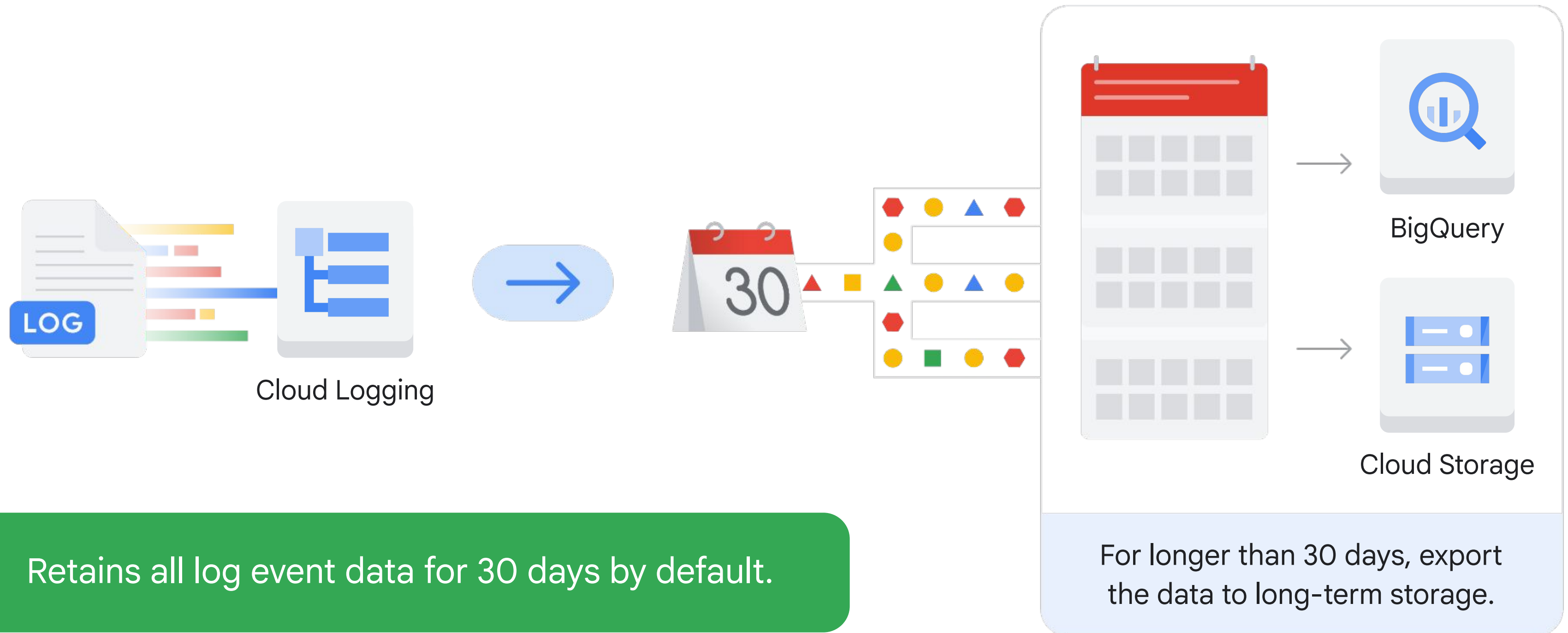


As the container runs, events happen and the log file grows.

The logrotate utility



All log events are streamed to Cloud Logging



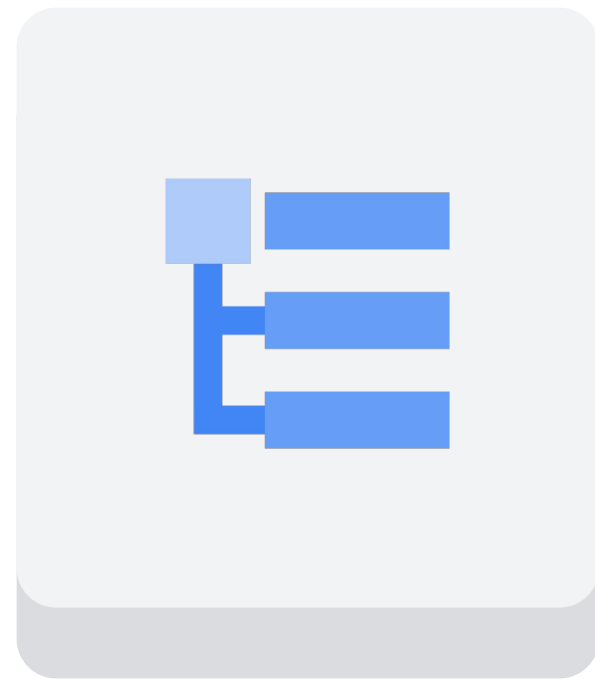
Retains all log event data for 30 days by default.

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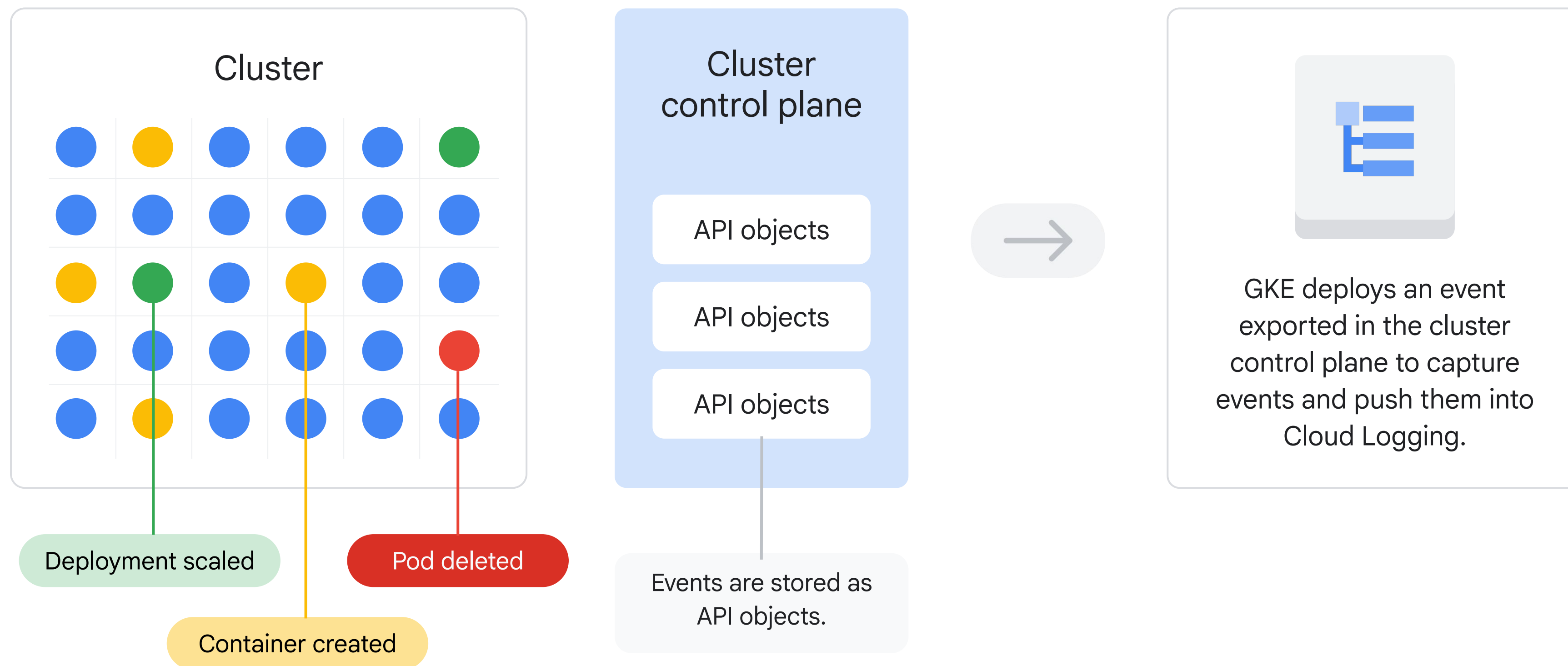
Cloud Logging



Cloud Logging

- Stores, searches, analyzes, monitors, and alerts on logging data and events from Google Cloud.
- Designed to automatically scale and ingest terabytes of log data per second.

Events are stored as API objects



GKE integration with Cloud Logging

Integration is enabled by default.



Can be disabled on a cluster.



Cloud Logging agents are:



Preinstalled on nodes.



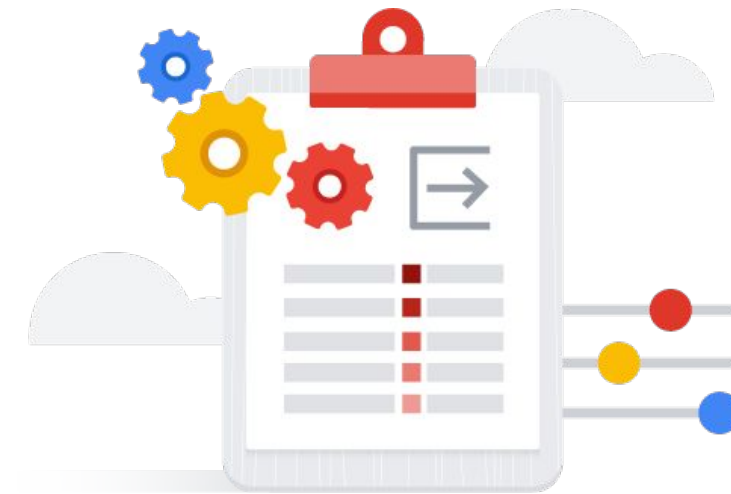
Pre-configured to export log data to Cloud Logging.

Custom log or filter logs



Cloud Logging API

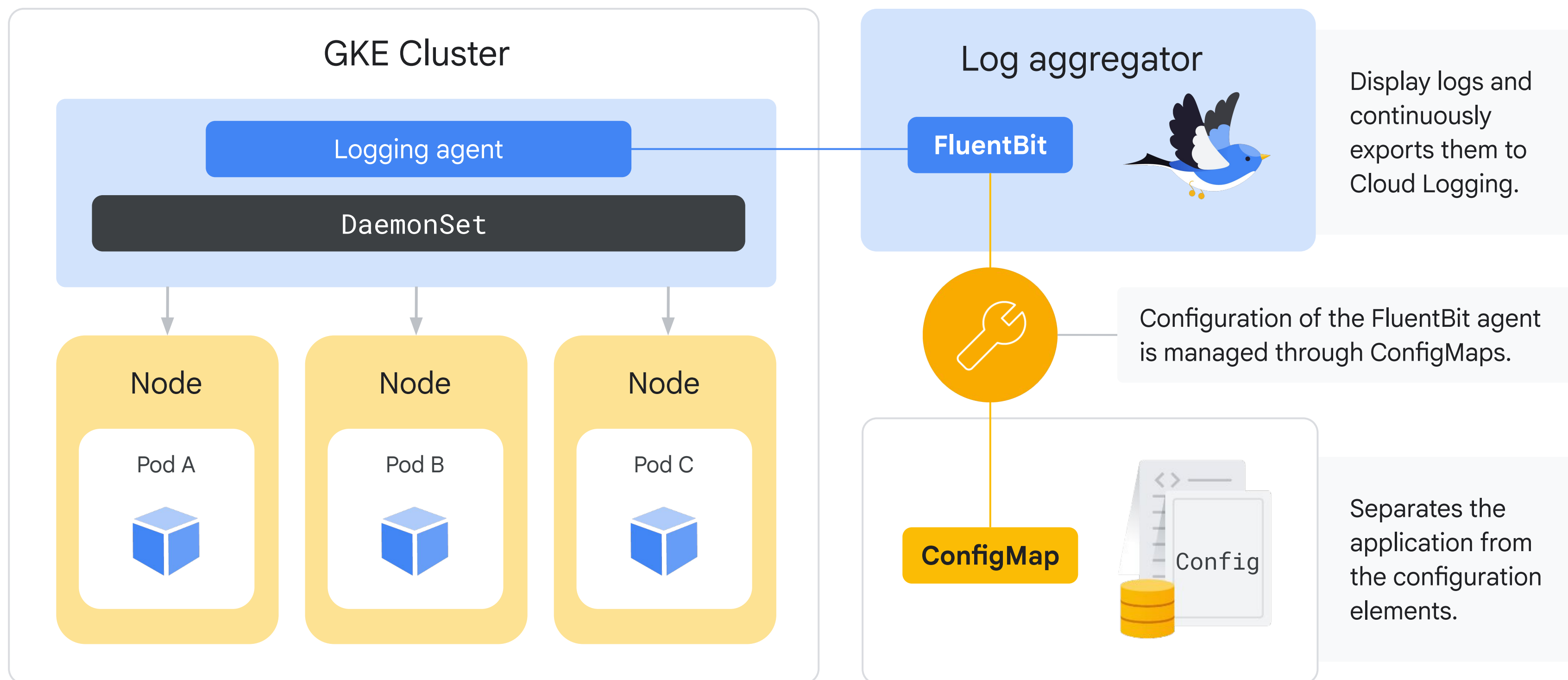
Write a custom log and push it
to Cloud Logging



Filter logs using Cloud filter language

- ☒ Logs Viewer console
- ☒ Cloud Logging API

FluentBit

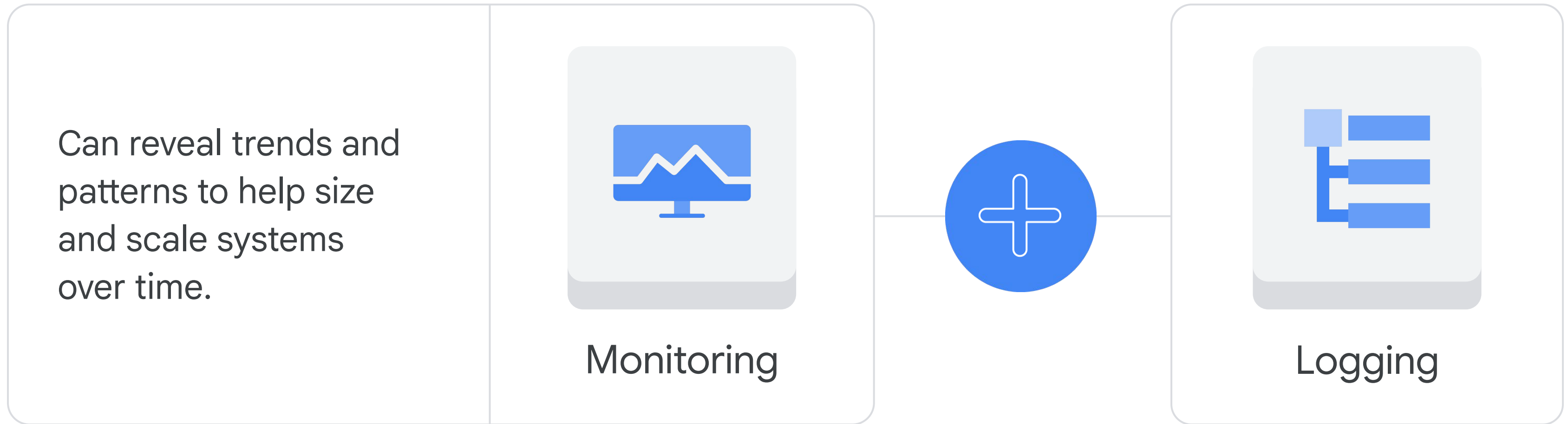


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Why does monitoring matter?



Monitoring and logging are often coupled to provide a complete picture of the system status and past trends.

Monitoring Kubernetes

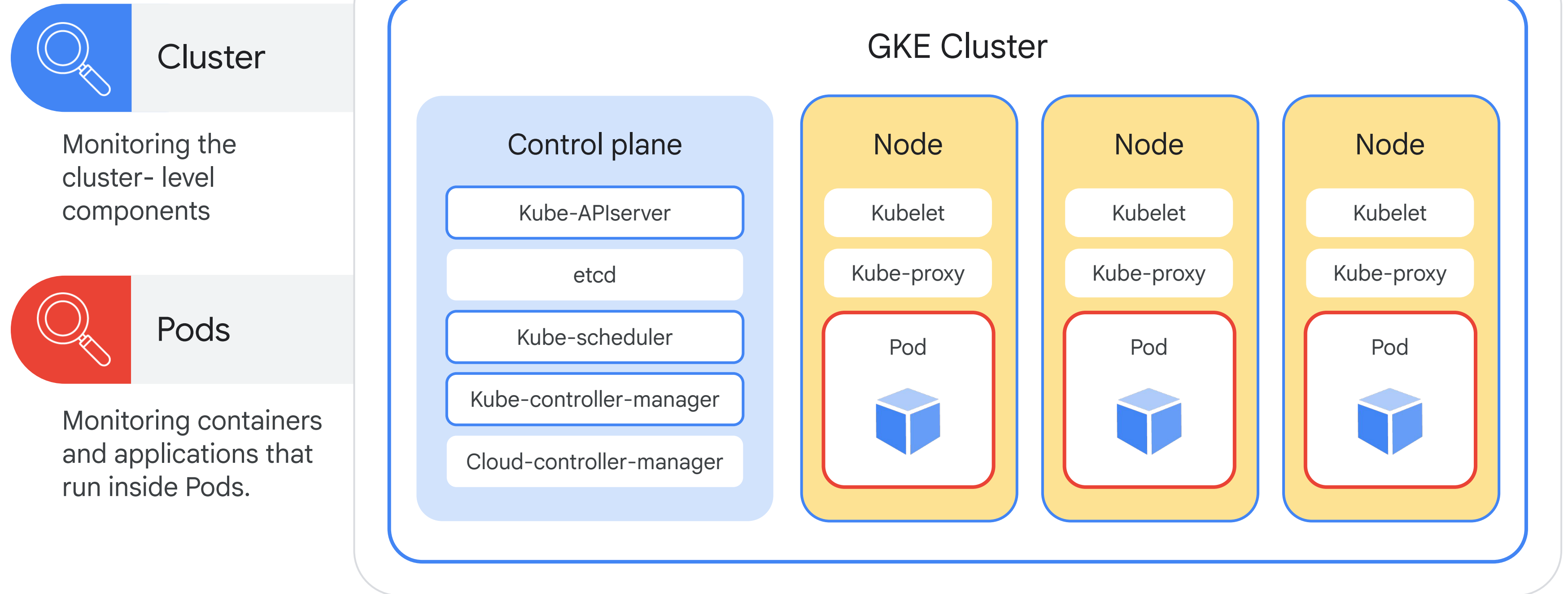


Monitor your application's current state and anything that might be impeding it.



Identify performance bottlenecks by accurately monitoring the state of services like the throughput and latency.

Kubernetes monitoring falls into two domains



Monitoring the cluster



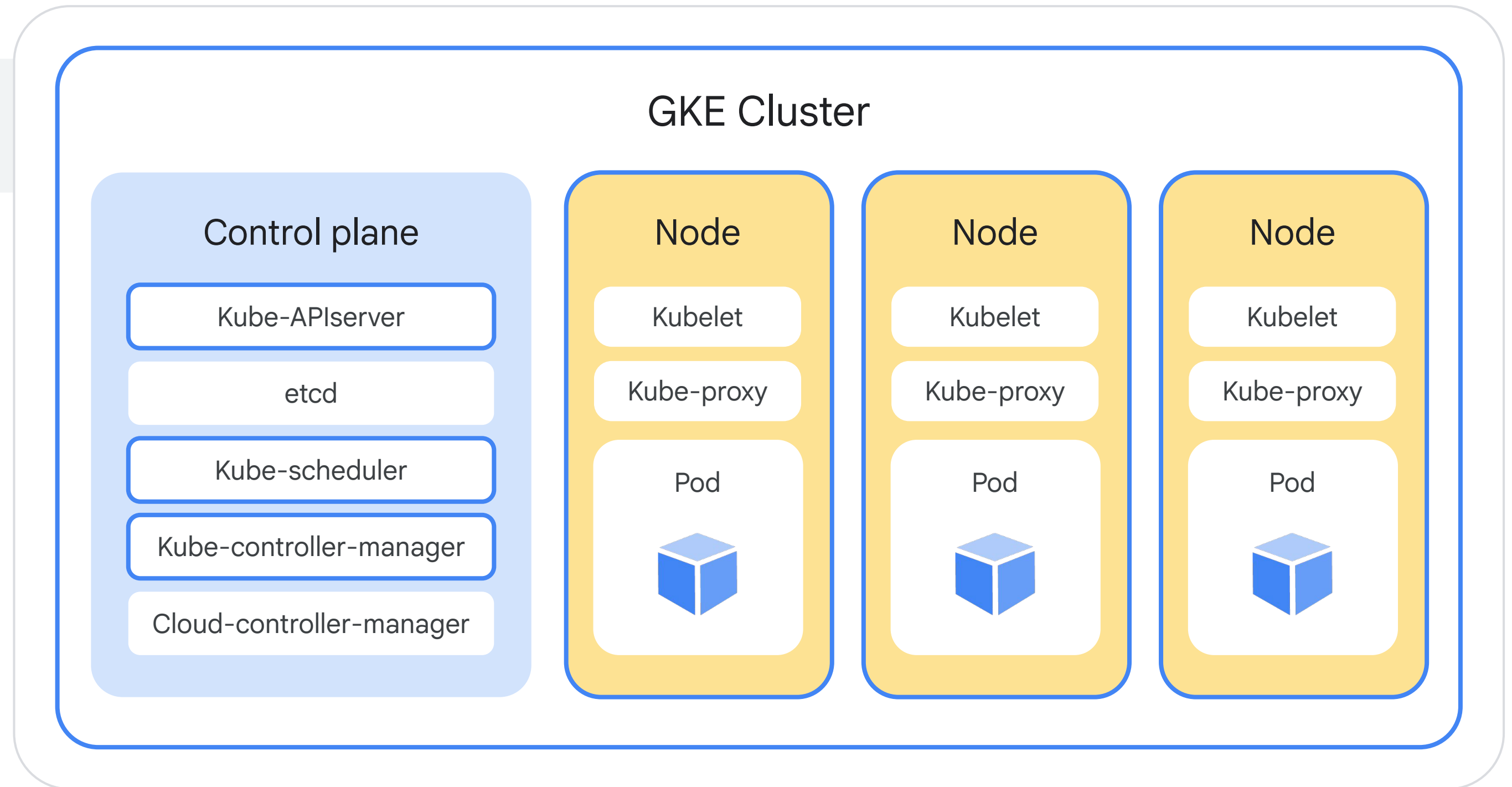
Cluster

Cluster monitoring refers to the cluster services, nodes, and other infrastructure elements.

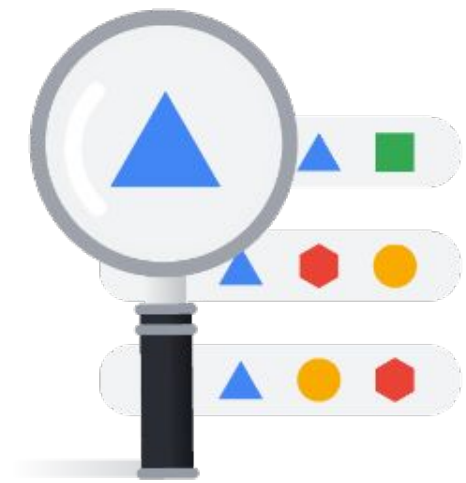
This can be accomplished using:

Google Cloud console

Cloud Monitoring



Events and metrics



Metrics

A value that can be monitored, such as CPU or disk usage.

Metrics report numerical values.



Events

Represent occurrences that happen to a cluster, Pod, or container.

Events report successes, warnings, or failures.

Pod monitoring metric sub-categories



System metrics

Container deployments
Instances
Health checks
State

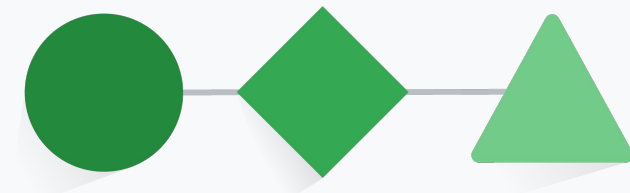


Container-specific metrics

Resource consumption



Prometheus



Application-specific metrics

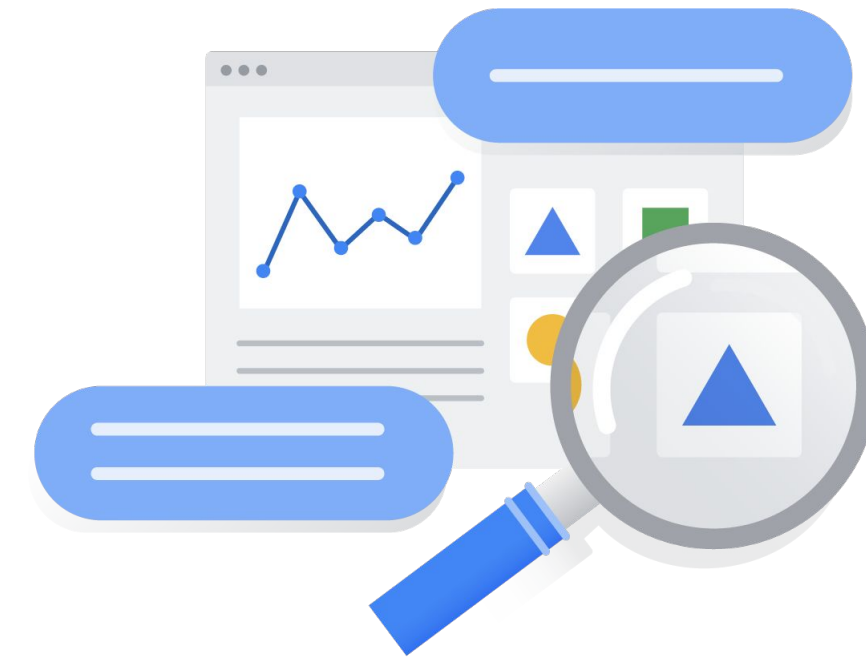
Metrics designed by the
application developer and
exposed to a monitoring
solution

Kubernetes monitoring requires a different approach



Traditional monitoring

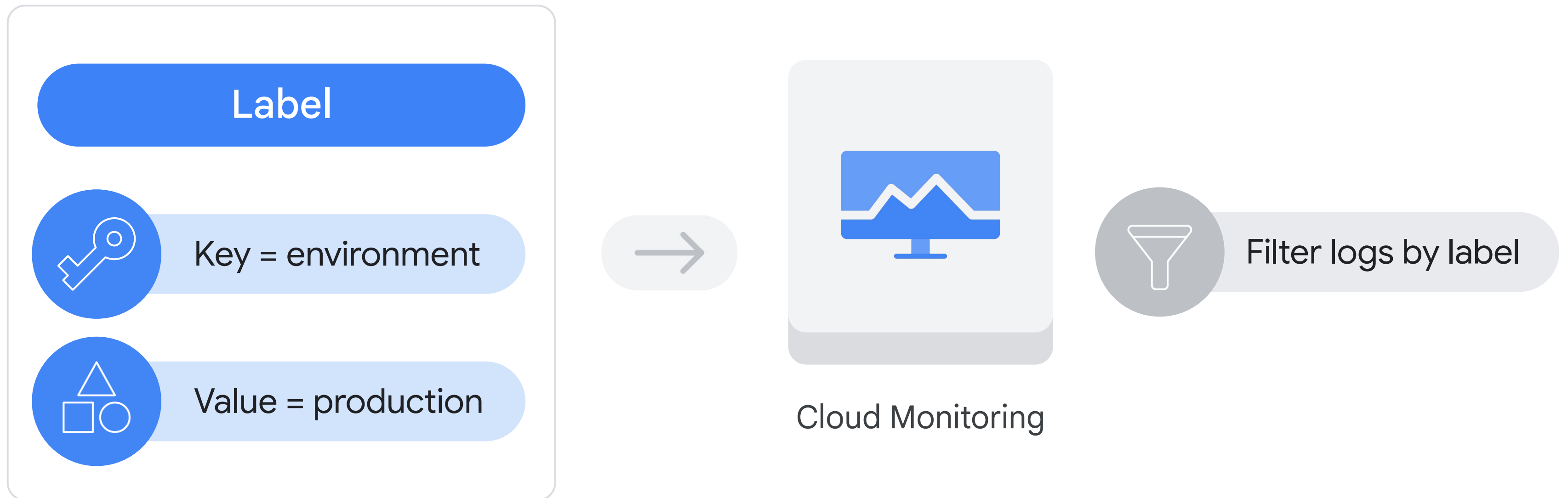
A specific hostname or IP address is monitored.



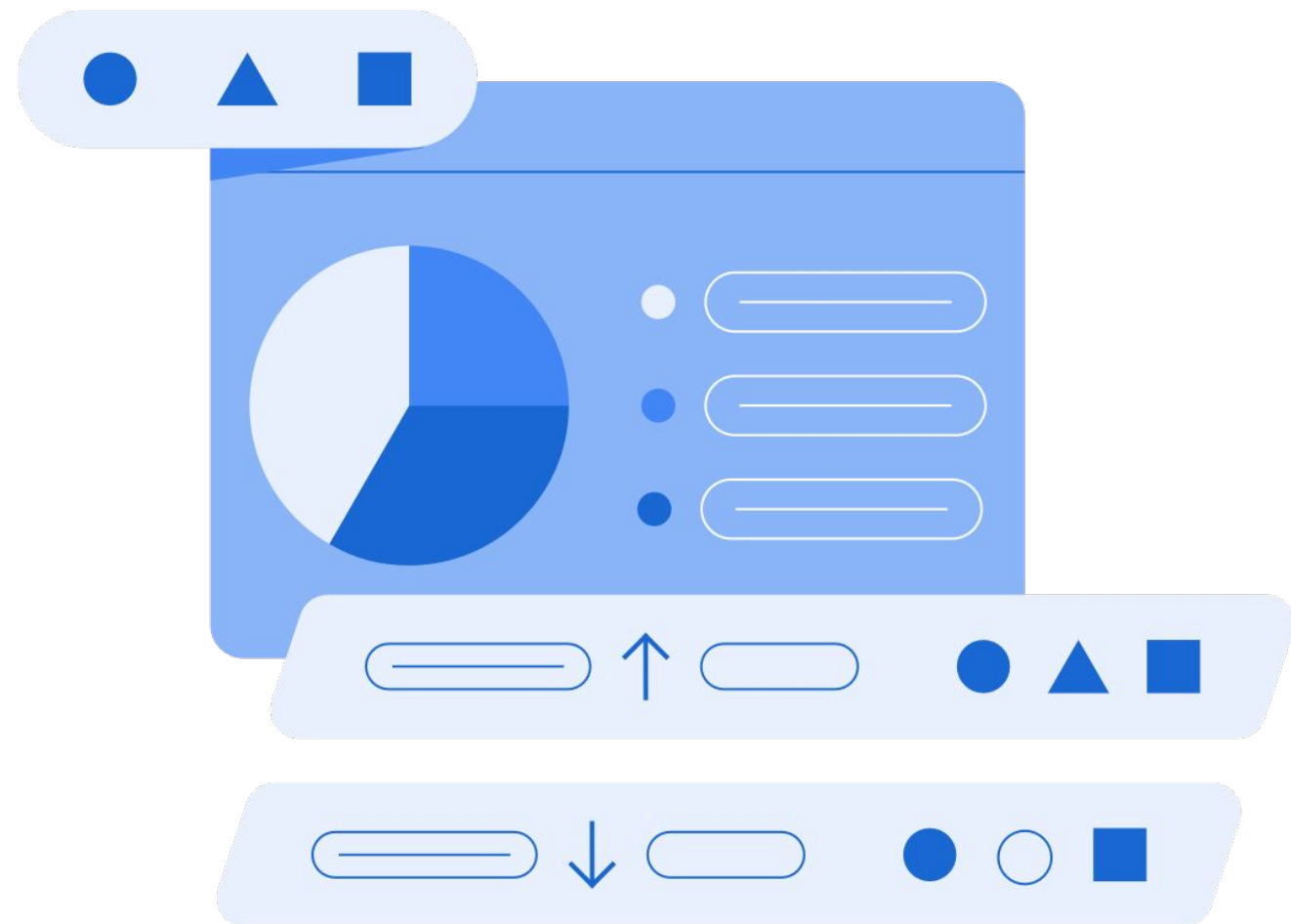
Kubernetes monitoring

Labels provide a logical approach to organizing resources, and make it easier to monitor specific systems or subsystems.

Cloud Monitoring can filter log-based metrics by label



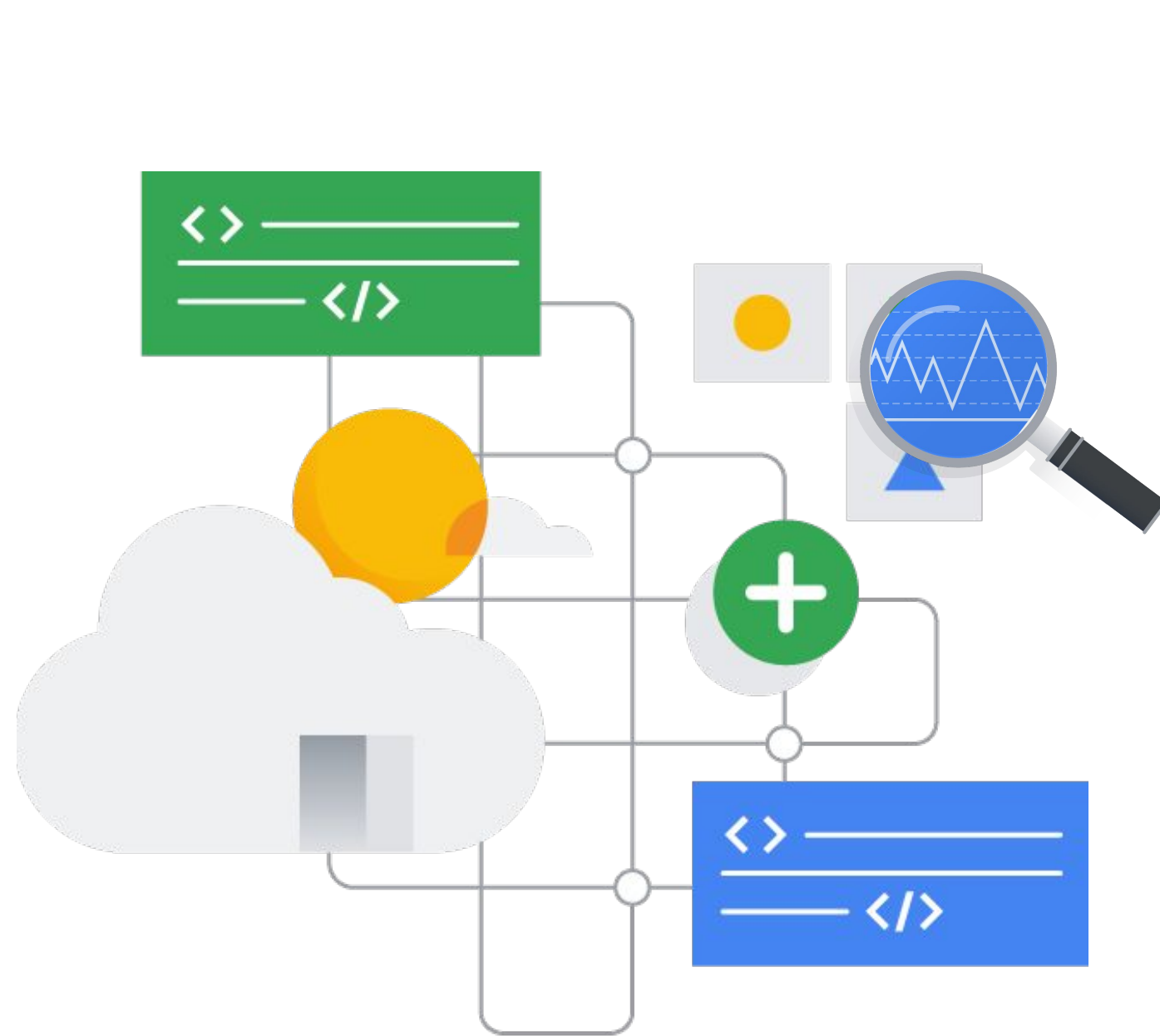
Dashboards provide cluster health insights



Key features include:

- ✓ Uptime checks to monitor cluster resources, control plane metrics, and Google Cloud resources.
- ✓ Integration with Cloud Logging to leverage custom metrics and monitor your logging-based custom metrics.
- ✓ Custom metrics for alerting policies, which can improve the signal-to-noise ratio of your alert messages.

Cloud Monitoring is optimized for Kubernetes



View metrics, logs, traces, events, and metadata in one place.



Analyze infrastructure metrics, application metrics, or logs from a single dashboard.



Supports multi-cluster monitoring within the Google Cloud environment, other cloud environments, and on-premises Kubernetes environments.



Offers extended direct support to Prometheus.

Cloud Monitoring + Prometheus

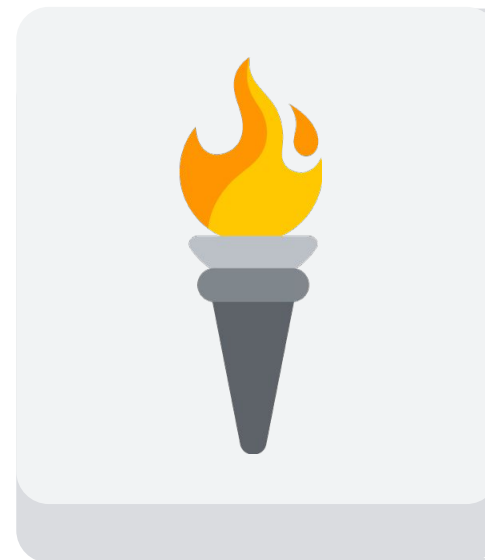
Cloud
Monitoring



Can only monitor
what it can see.

Direct integration

Google Cloud Managed
Service for Prometheus



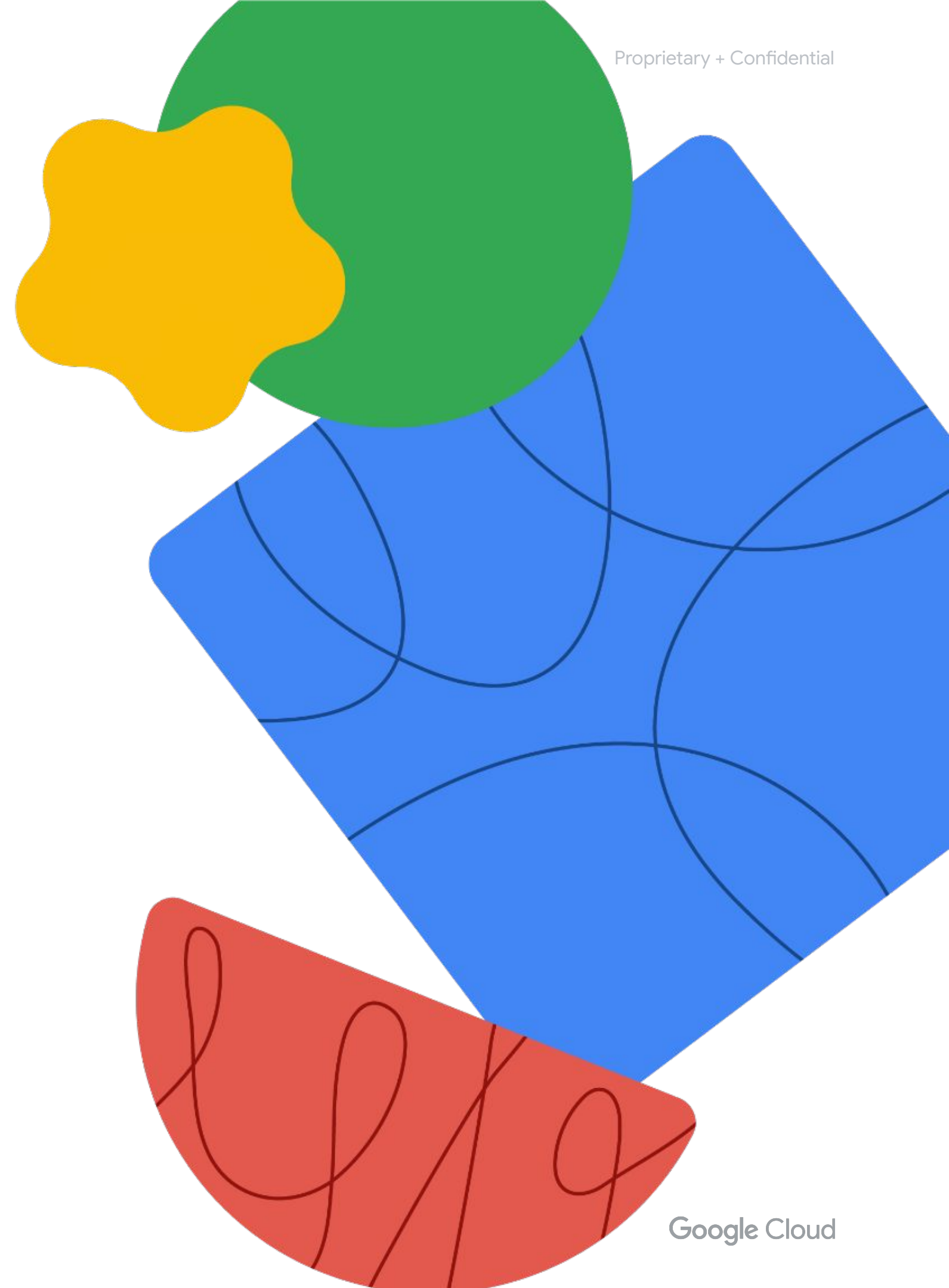
Open-source monitoring and
alerting toolkit that collects and
stores time-series data.

Provides detailed metrics about
Kubernetes components.

Exposes those metrics
to Cloud Monitoring.

Quiz questions

Let's pause for a quick check in.



Quiz | Question 1

Question

You are a systems administrator tasked with monitoring and maintaining a cluster of Google Kubernetes Engine (GKE) instances. Which tool would you use to collect and store logs from your GKE cluster?

- A. Cloud Logging
- B. Google Kubernetes Engine
- C. Error Reporting
- D. Kubernetes

Quiz | Question 1

Answer

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Quiz | Question 2

Question

You need to monitor specific applications running on your production GKE Clusters. How can you create a logical structure for your application that allows you to selectively monitor the application's components using Cloud Monitoring? Choose all responses that are correct (2 correct responses).

- A. Filter the Cloud Monitoring logs using the application name.
- B. Filter the Cloud Monitoring logs using Kubernetes labels.
- C. Add a prefix to all Pod names that identify the application.
- D. Add Labels to the Pods in Kubernetes that identify the applications.
- E. Filter the Cloud Monitoring logs using the application prefix.

Quiz | Question 2

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Quiz | Question 3

Question

You are troubleshooting an issue which happened in the last hour. You enter the command 'kubectl logs --since=3h demo-pod'. However, the events you are looking for do not appear in the output. What is the likely cause?

- A. The file has been archived to Cloud Logging and is no longer available locally.
- B. The log file contains more than 3 hours of data, and it has been archived.
- C. The log file is older than 2 hours and is not included in the results.
- D. The log file was greater than 100MB in size, and it has been rotated.

Quiz | Question 3

Answer

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Quiz | Question 4

Question

Which component is specifically responsible for collecting logs on a GKE node and forwarding them to Cloud Logging?

- A. DaemonSet
- B. FluentBit
- C. Cloud Logging
- D. Kubernetes

Quiz | Question 4

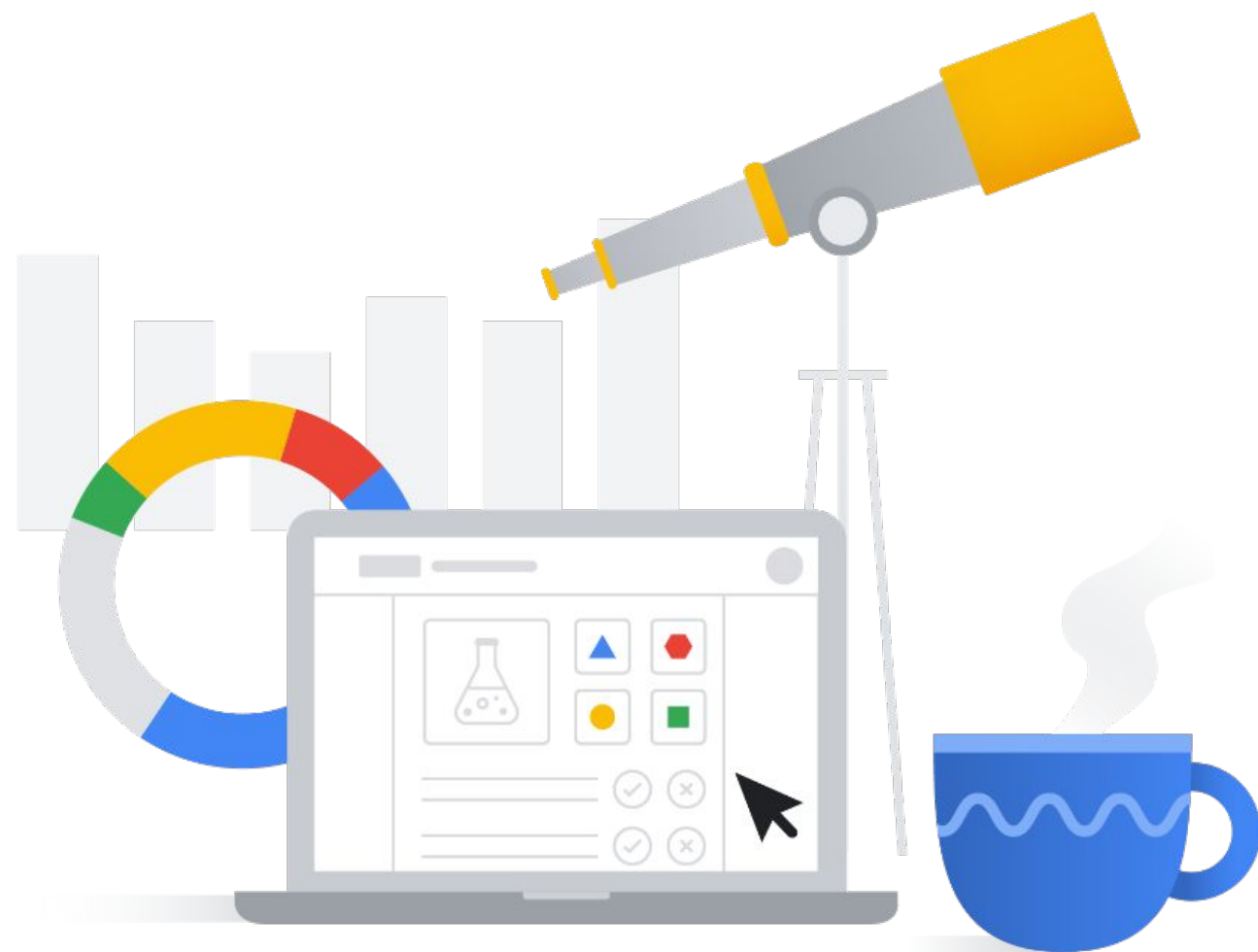
Answer

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Lab: Configuring GKE-Native Monitoring and Logging



01

Use Kubernetes Engine Monitoring to view cluster and workload metrics.

02

Use Cloud Monitoring alerting to receive notifications about the cluster's health.